

CADRE AI

AI Engineer & FDE Technical Take-Home

Chatbot Challenge | Candidate Guide | v1.1

We don't test what you've memorized. We test how fast you can build.

Read this guide before starting. It covers exactly what to expect, what to deliver, and how you'll be evaluated.

cadreai.com

Welcome

Thanks for taking the time to work through this challenge. We're excited to see what you can build.

Our engineers build production AI systems every day using Claude Code as their primary development tool. We don't believe LeetCode tells us anything useful about whether you'll thrive in that environment. Instead, we're asking you to do what our team does daily: take a product brief, break it down, make deliberate decisions, build it with AI-augmented tooling, and ship it.

This guide tells you everything you need to know. There are no trick questions and no gotchas. We want you to perform at your best.

Challenge Format

This is an async take-home challenge followed by a 1-hour live review with your interviewer.

Phase	What Happens
The Build	Complete the challenge on your own time before your scheduled review. We recommend budgeting 4–6 hours. Build, deploy, and prepare your submission.
The Review	A 1-hour live conversation. You'll demo your deployed app, walk through your architecture and decisions, and discuss your AI workflow. A conversation, not a quiz.

 Submit your completed challenge — with a live deployed URL — at least one full business day before your scheduled review. This gives your interviewer time to review your project before the session.

 We recommend spending 4–6 hours on the build. There's no hard time limit — quality and judgment matter more than speed — but a strong senior engineer should be able to ship a solid MVP in that window. Don't spend your whole weekend on it.

Your Challenge

The Brief

You have been brought in to build a customer support chatbot for Cadre AI.

Cadre AI is an AI strategy and implementation consultancy. We help businesses move from AI confusion to AI confidence — going department by department to identify high-ROI AI opportunities, build workflows and agents, and train teams so the changes actually stick. Our clients range from lower middle market private equity-backed companies to professional services firms and financial services organizations.

Cadre's inbound team is receiving a growing volume of inquiries from prospective clients, existing clients, and people who want to learn more about what we do. Your job is to build a chatbot that handles the most common interactions so the team can focus on high-value conversations.

About Cadre AI

Item	Detail
What we do	AI strategy, workflow automation, AI agents, and leadership facilitation for businesses
Who we serve	B2B companies: professional services, private equity, financial services, real estate, construction, manufacturing, retail, and more
Core services	AI Strategy, AI Leadership & Facilitation, AI Engineering, AI Agents
Key partners	OpenAI, Anthropic (Claude), Google, Microsoft, AWS, Salesforce, Snowflake — plus OpenRouter for model access
Common inquiries	What does Cadre AI do, how to get started, service pricing, AI Maturity Index, case studies, how to book a strategy call, what industries we work in

What to Build

Build a customer support chatbot for Cadre AI that can handle common inbound inquiries. The prompt is intentionally underspecified — how you scope and prioritize is part of the evaluation.



Minimum bar: a functional chatbot that a prospective or existing Cadre AI client could plausibly use to get answers. Everything beyond that is your call.

Some scenarios the bot should be able to handle (treat this as a starting point, not an exhaustive spec):

- A prospective client asking what Cadre AI does and whether we work with their industry
- Someone asking how to book a call with an AI strategist
- A client asking how to access the Cadre portal to track their AI tools, agents, and results
- A business leader asking what the AI Maturity Index is and how to get scored
- Someone asking about Cadre's approach to LLM selection and data security
- A user asking a question the bot can't answer — and needs to escalate or redirect



You decide what the bot knows. You decide where it draws the line. You decide what's in scope. We're watching those decisions closely.

Deliverables

Before your review session, make sure you have the following ready:

- A deployed, publicly accessible URL of your chatbot
- Upload a zip of your complete project through the submission link provided by our recruiting team
- Exclude dependency folders like `node_modules`, `dist`, `build`, or your virtual environment before zipping. Your project's source code should be lightweight, typically just a few megabytes at most.
- A `CLAUDE.md` at the root of the project
- A `plan.md` at the root of the project

 Include your `.git` folder in the zip. This is how we review your commit history, pacing, and methodology, so don't exclude it along with the other dependency folders.

 Create a `CLAUDE.md` and `plan.md` at the root of your project. Use git to commit your code as you build. When you're ready to submit, zip your project (excluding dependency folders) and upload it through the submission link. The app must be deployed and accessible on a public URL. You will walk through your code, architecture, and Claude Code workflow in the review.

What We're Looking For

We evaluate five dimensions, weighted by importance to Cadre AI's engineering culture:

Dimension	Weight	In Practice
Claude Code Proficiency	30%	How you set up CLAUDE.md, plan.md, use subagents, custom commands, and manage AI context. The most important dimension.
System Design & Architecture	25%	Your data model, API structure, system prompt design, separation of concerns, and scaling trade-offs.
Development Speed & Scope	20%	How you prioritize features, where you draw scope boundaries, and how you manage complexity.
Code Quality & Verification	15%	Clean code, error handling, catching AI bugs, knowing what your code does.
Communication & Reasoning	10%	Explaining decisions, articulating trade-offs, productive technical dialogue.



We're not testing whether you can code. We're testing whether you can think clearly enough to direct and verify a system that codes with you.

How to Prepare

1. Get Comfortable with Claude Code

If you haven't used Claude Code before, spend time with it first. Key things to practice:

- Running `/init` on a new project and customizing the generated CLAUDE.md
- Writing a plan.md that breaks a project into phases Claude can execute sequentially
- Using subagents to parallelize independent tasks
- Debugging when Claude generates broken code — providing error output and context back to Claude
- Knowing when to accept Claude's output, modify it, or reject it entirely

2. Have a Go-To Tech Stack

Speed matters. Pick what you know. There is no required tech stack — the only requirement is that it deploys and works on a public URL.

Stack	Strengths	Good For
Next.js + Supabase	Auth built-in, fast Vercel deploys	Great default choice
Python + FastAPI + React	Flexible, great for AI integrations	AI-heavy apps
T3 Stack	Full type safety, tRPC	Complex UIs, real-time
Rails / Django	Batteries included, fast scaffolding	Rapid prototyping

3. Practice the Full Loop

The review tests the complete cycle: plan → build → deploy → iterate. Practice building a small app end-to-end using Claude Code before the challenge. Your goal is a working, deployed MVP without scrambling.

4. Know Your CLAUDE.md Strategy

During the review we'll ask about your CLAUDE.md. Think of it as onboarding documentation for an extremely fast but context-limited junior developer. The best CLAUDE.md files are opinionated and specific — not generic boilerplate.

What Happens in the Review

After you submit, we'll have a 1-hour live conversation:

Section	~Time	What We'll Discuss
Live Demo	10 min	Walk through the deployed app. Show what works, be upfront about what's broken or out of scope.
Architecture	15 min	System prompt design, API structure, data model, scaling considerations.
Claude Code Workflow	15 min	CLAUDE.md, plan.md, subagents, prompting strategy, how you handled AI errors.
Code Deep Dive	10 min	Specific functions explained. What Claude generated vs. what you modified, and why.
Decisions & Trade-offs	10 min	Scope choices, what you left out intentionally, what you'd do with more time.

✓ We expect trade-offs. We expect incomplete features. The best candidates are honest about what's broken and articulate about what they'd do with more time.

Tips from the Cadre AI Engineering Team

✓ Do This	✗ Avoid This
Plan before coding. Write CLAUDE.md and plan.md first.	Starting to code immediately with no plan or context.
Deploy early. Get something live before you start iterating.	Waiting until the end — deployment issues are real.
Small, frequent commits with descriptive messages.	One giant commit at the end.
Read and verify Claude's output. Test as you go.	Blindly accepting everything Claude generates.
Use subagents for independent tasks.	One massive prompt, long wait, wall of unverified code.
Cut scope aggressively. 3 working features > 8 broken ones.	Trying to build everything and shipping nothing.
Give Claude error messages + context when debugging.	Re-running the same failing prompt repeatedly.
Make your scope decisions explicit in plan.md.	Leaving scope ambiguous and hoping for the best.

Frequently Asked Questions

Can I use other AI tools besides Claude Code?

Claude Code is the primary tool we evaluate on. You can use the internet freely, but lean into Claude Code — it's 30% of your score.

What if I've never used Claude Code?

Spend several hours practicing before the challenge. It's free to install. Familiarity matters — it's the highest-weighted dimension.

Which LLM should I use for the chatbot?

That's part of the challenge. You have full flexibility on model selection. Choose what you think is right for the job and be ready to explain why during the review.

What if my deployment breaks?

Deployment issues happen. How you handle them is part of the evaluation. Your interviewer can help with platform configuration but not code issues.

What if I don't finish everything?

Nobody builds everything. A focused MVP with clear trade-offs beats an ambitious mess every time. Be explicit about what you left out and why.

Can I use component libraries and third-party packages?

Yes. Use whatever helps you move fast and build well. We're testing engineering judgment and AI workflow, not your ability to write CSS from scratch.

Good luck. Build something great.

We're rooting for you.

— The Cadre AI Engineering Team